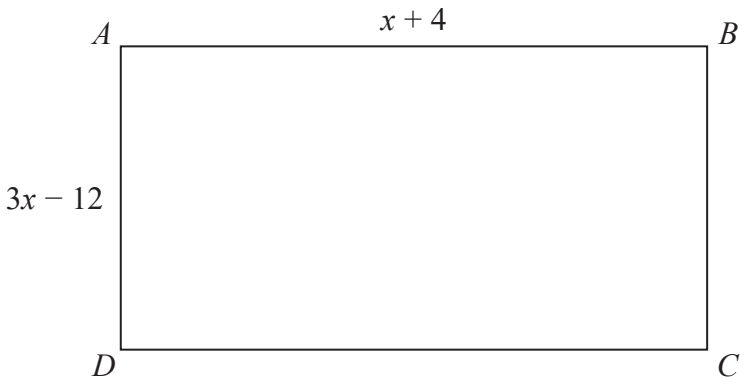
21 *ABCD* is a rectangle.



All measurements are in centimetres.

The perimeter of the rectangle is 38 cm.

(a) Work out the length of *AD*.

$2[(x + 4) + (3x - 12)] = 38$

1 mark
Form perimeter equation.

$4x - 8 = 19$

$4x = 27$, so $x = 6.75$

1 mark
Collect x terms.

$AD = 3 \times 6.75 - 12 = 8.25$ cm

1 mark
Substitute into AD.

1 mark
AD = 8.25 cm.

..... cm

(4)

Jamal says,

"If I double the value of x then the perimeter of the rectangle will double."

(b) Is Jamal correct?
You must give a reason for your answer.

No. $P = 8x - 16$, so doubling x gives $16x - 16$,

whereas doubling P gives $16x - 32$. These differ.

1 mark
The constant term does not double.

(1)

(Total for Question 21 is 5 marks)

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